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SEPTEMBER 4, 1958
SUPERSEDING
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September 13, 1946

FEDERAL SPECIFICATION

BAND, LIGATURE, DENTAL, RUBBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers dental ligature rubber bands suitable for orthodontic use and as an accessory part of a splint fracture set.

1.2 Classification.

1.2.1 *Sizes.*—Dental ligature bands shall be of the following sizes, as specified (see 6.1):

Size 1.—Large.
Size 2.—Small.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Test Method Std. No. 601—Rubber; Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-B-10377—Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL-L-10547 — Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The bands shall be made of pale crepe rubber.

3.2 Physical requirements.—Bands shall be seamless. The tensile strength of the bands shall be not less than 1300 pounds per square inch, and the ultimate elongation shall be not less than 700 percent, when tested as specified in 4.3. After being subjected to the seven day accelerated aging test specified in 4.3, the bands shall show a loss in tensile strength of not more than 20 percent, and a loss in ultimate elongation of not more than 10 percent, from the original value.

3.3 Dimensions.—Dimensions of the bands shall be as shown in table I, when tested as specified in 4.3. A plus or minus tolerance of 0.005 inch will be permitted for thickness and width, and a plus or minus tolerance of $\frac{1}{32}$ inch will be permitted for the outside diameter of the band.

TABLE I.—*Dimensions of bands*

Size	Thickness	Width	Outside Diameter
	Inch	*Inch*	*Inch*
1	0.031	0.062	$\frac{3}{8}$
2	0.031	0.062	$\frac{1}{4}$

3.4 Workmanship.—Workmanship shall be first class throughout. The bands shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *For visual examination.*—Sampling procedures for visual examination shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.1.2 *For tests.*—Sample size for each test shall be as indicated for inspection level L–2 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero for all sample sizes, and a minimum of two samples shall be taken.

4.2 Inspection.—The inspector shall ascertain that the dental ligature bands have been examined and tested in accordance with specification requirements.

4.3 Tests.—Tests shall be performed for the requirements specified in 3.2 and 3.3, and shall be conducted in accordance with Federal Test Method Standard 601, for the test method specified in table II.

TABLE II.—*Test methods*

Test	Requirements Paragraph	Test Method
Tensile strength	3.2	4111[1]
Ultimate elongation	3.2	4121[1]
Accelerated aging	3.2	7111[2]
Thickness	3.3	2031
Width	3.3	2111
Outside diameter	3.3	2221

[1] Except each grip shall consist of one roller with a maximum diameter of $\frac{1}{8}$ inch.
[2] Except 7-day aging.

4.4 Supplier responsibilities for inspection.—The supplier shall perform examinations and tests set forth in this specification to substantiate conformance of supplies to specification requirements. The examinations and tests shall be at the expense of the supplier. Suppliers shall have adequate test facilities or shall arrange for test facilities acceptable to the Government to be used for executing specific prescribed tests. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government upon the Government's request, at any time, or from time

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to time, for a period of 2 years after the delivery of the supplies to which these records relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

5. PREPARATION FOR DELIVERY

The levels of preservation, packaging, and packing protection used are defined in Federal Standard No. 102 (for civil agencies only).

5.1 Packaging.

5.1.1 *Unit of issue.*—One hundred seventy-five bands, size 1, or two hundred twenty-five bands, size 2, as specified, constitute one unit of issue.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table III shall be packaged in each unit, intermediate and exterior container as applicable for the required level of protection specified in procurement document.

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size and design, or a complete wrapping with kraft paper of suitable weight. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or Military Specification MIL-B-10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

TABLE III.—*Packaging quantities*

Federal Stock Number	Size	Unit package	Intermediate package	Shipping container
6520-501-5323	1	1	24	288
6520-501-5325	2	1	24	288

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5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Unit package.*—Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.2 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.3 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Size required (see 1.2.1).
(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Armed Services Medical Stock List:

6520–501–5323 BAND, LIGATURE, DENTAL, Rubber, Large, 175s.
6520–501–5325 BAND, LIGATURE, DENTAL, Rubber, Small, 225s.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Navy—Md
Army—M
Air Force.

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